



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE (Advanced)

FULL SYLLABUS

08-05-2024

JEE(Main + Advanced) : ENTHUSIAST COURSE ALL STAR BATCH (SCORE ADVANCED)

ANSWER KEY

PAPER-1 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I(i)	Q.	1	2	3	4	5	6
	A.	C	D	B	C	C	B
SECTION-I(ii)	Q.	7	8	9	10	11	12
	A.	B,D	A,C	A,B,C,D	B,D	A,C	A,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	8.00	2.00	5.00	8.00	2.00	17.00

PART-2 : CHEMISTRY

SECTION-I(i)	Q.	1	2	3	4	5	6
	A.	C	C	A	B	C	A
SECTION-I(ii)	Q.	7	8	9	10	11	12
	A.	A,B,C	B,D	B,C,D	A,B,C	A,D	A,B,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	3.00	4.50	3.00	6.00	556.10 to 557.80	8.90 to 9.10

PART-3 : MATHEMATICS

SECTION-I(i)	Q.	1	2	3	4	5	6
	A.	D	C	D	A	B	B
SECTION-I(ii)	Q.	7	8	9	10	11	12
	A.	B,C,D	A,C,D	A,B,D	C,D	A,B,C	B,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	9.50	119.00	0.50	1206.00	19.20	5.00

HINT – SHEET

PART-1 : PHYSICS

SECTION-I(i)

1. Ans (C)

$$A_r = \frac{\sqrt{\mu_1} - \sqrt{\mu_2}}{\sqrt{\mu_1} + \sqrt{\mu_2}} A_i$$

$$A_t = \frac{2\sqrt{\mu_1}}{\sqrt{\mu_1} + \sqrt{\mu_2}} A_i$$

$$\Rightarrow \frac{A_r}{A_t} = -\frac{1}{2} \text{ and } P = \frac{1}{2} \mu v \omega^2 A^2$$

$$= \frac{1}{2} \mu \sqrt{\frac{T}{\mu}} v \omega^2 A^2 \quad \left(\because v = \sqrt{\frac{T}{\mu}} \right)$$

$$P \propto \sqrt{\mu} A^2$$

$$\left[\frac{P_t}{P_r} = \sqrt{\frac{\mu_t}{\mu_r}} \left(\frac{A_t}{A_r} \right)^2 \right]$$

$$= \sqrt{\frac{4\mu}{\mu}} (2)^2 = 8$$

2. Ans (D)

$$2R = \frac{3R}{2} + \frac{PdV}{dT}$$

$$\frac{PdV}{dT} = \frac{R}{2} \Rightarrow \frac{RT}{V} \frac{dV}{dT} = \frac{R}{2}$$

$$\Rightarrow \int \frac{dV}{V} = \frac{1}{2} \int \frac{dT}{T}$$

$$\ln V + \ln C = \frac{1}{2} \ln T \text{ (where C is constant)}$$

$$\ln CV = \frac{1}{2} \ln T \Rightarrow T = C^2 V^2$$

$$\text{Hence } T \propto V^2$$

$\therefore$ Temperature becomes four times.

3. **Ans (B)**

$$I_0 = \frac{1}{4} \times (4M) \frac{a^2}{6}$$

$$= \frac{Ma^2}{6} = \frac{7 \times 144}{6} = 168 \text{ kg cm}^2$$

4. **Ans (C)**

Lens formula gives

$$\frac{1}{v} - \frac{1}{(-100)} = \frac{1}{50} \Rightarrow v = 100 \text{ cm}$$

The mirror forms an image 50 cm behind itself and the image is moving to right with a velocity of 20 m/s. The image formed by the mirror will act as moving object for the lens.

Again applying the lens formula gives

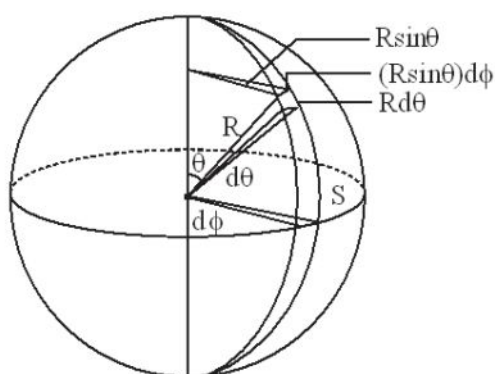
$$\frac{1}{v} - \frac{1}{(-200)} = \frac{1}{50} \Rightarrow v = \frac{200}{3} \text{ cm}$$

For lens

$$\frac{dv}{dt} = \left(\frac{v}{u}\right)^2 \frac{du}{dt} = \left(\frac{200/3}{200}\right)^2 \cdot 20 = \frac{20}{9} \text{ cm/s}$$

5. **Ans (C)**

The given charge distribution is equivalent to charge on a strip S on a uniformly charged spherical shell of radius R.



Since electric field in the sphere is zero, therefore net electric field due to any two diametrically opposite such strips on all points of the common diameter must be zero. It suggests that electric field of a single strip and hence electric field of the given semicircular arc anywhere on their diameter cannot have a component parallel to the diameter.

6. **Ans (B)**

$$\frac{dN}{dt} = t^2 - \lambda N$$

$$\text{For } \frac{dN}{dt} \text{ to be minimum } \frac{d^2N}{dt^2} = 0$$

$$\Rightarrow \frac{d^2N}{dt^2} = 2t - \lambda(t^2 - \lambda N) = 0$$

$$\text{Or } N = \frac{2t_0 - \lambda t_0^2}{\lambda^2}$$

PART-1 : PHYSICS

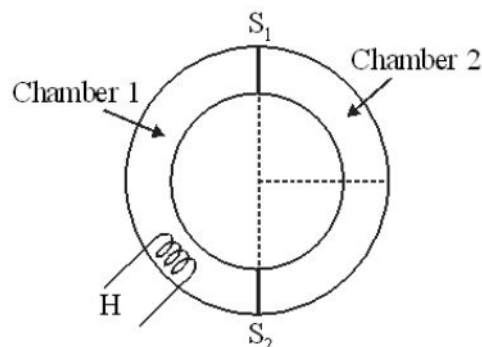
SECTION-I(ii)

7. **Ans (B,D)**

The process in chamber 2 is adiabatic.

Let final pressure in chamber 2 be P_2 .

$$P_2 \left(\frac{V_0}{2}\right)^\gamma = P_0 V_0^\gamma$$



$$\Rightarrow P_2 = (2)^{3/2} P_0 \left[\because \gamma = \frac{3}{2} \right]$$

$$= 2\sqrt{2} P_0$$

In equilibrium, pressure in both chambers must be same, hence final pressure in chamber 1 is

$$P_1 = P_2 = 2\sqrt{2} P_0$$

Work done on the gas in chamber 2 by gas in chamber 1 is

$$W = \frac{P_2 V_2 - P_0 V_0}{\gamma - 1} = \frac{2\sqrt{2} P_0 \frac{V_0}{2} - P_0 V_0}{\frac{3}{2} - 1}$$

$$W = 2(\sqrt{2} - 1) P_0 V_0$$

Change in internal energy of the gas in chamber 1

$$\Delta U = n C_v \Delta T = \frac{n R \Delta T}{\gamma - 1} = 2n R \Delta T$$

$$= 2[P_1 V_1 - P_0 V_0] = 2 \left[2\sqrt{2} P_0 \cdot \frac{3}{2} V_0 - P_0 V_0 \right]$$

$$= 2(3\sqrt{2} - 1) P_0 V_0$$

$$\therefore \Delta Q = \Delta U + W = 2(3\sqrt{2} - 1) P_0 V_0 + 2(\sqrt{2} - 1) P_0 V_0$$

$$= 4P_0 V_0 [2\sqrt{2} - 1]$$

8. **Ans (A,C)**

Loss in electrostatic potential energy

$$U_i - U_f = \lambda E_0 \frac{L^2}{2}$$

This is equal to gain in gravitational potential energy

$$\lambda E_0 \frac{L^2}{2} = Mg \frac{L}{2} \Rightarrow E_0 = \frac{Mg}{\lambda L}$$

The electric force on the rod is $F_e = \lambda L E_0$.

Force can be considered at centre for writing the torque.

In equilibrium,

$$\tau_{Fe} = \tau_{mg} \text{ [torque about hinge]}$$

$$\lambda L E_0 \frac{L}{2} \cos \theta = Mg \frac{L}{2} \sin \theta$$

$$\text{Putting } E_0 = \frac{Mg}{\lambda L} \text{ we get } \theta = 45^\circ$$

The rod has maximum KE at this position.

9. **Ans (A,B,C,D)**

$$mg \left(h + \frac{H}{2} \right) = [\nu \rho g] \frac{H}{2}$$

$$h = H$$

$$mg \left(h + \frac{H}{2} + \frac{H}{2} \right) = [\nu \rho g] \frac{H}{2} + [\nu 2 \rho g] \frac{H}{2}$$

$$h = \frac{7H}{2}$$

$$mg \left(h + \frac{H}{2} \right) - [\nu \rho g] \frac{H}{2} = \text{K.E.}$$

$$mg \left(\frac{5H}{2} \right) - \frac{3}{2} mgH = \text{K.E.}$$

$$\text{K.E.} = mgH.$$

10. **Ans (B,D)**

$$W_g + W_{fr} = \Delta K$$

W_g and ΔK same for all cases so W_{fr} should also be same

$$N_1 < N_2 \text{ and } \ell_1 < \ell_2 \Rightarrow \mu_1 > \mu_2$$

[N = Normal forces, ℓ = length of the path]

$$N_2 > N_3 \text{ and } \ell_2 = \ell_3 \Rightarrow \mu_2 < \mu_3$$

11. **Ans (A,C)**

For $r = 0$

$$R_{eq} = 1.5 R$$

for $r = \infty$

$$R_{eq} = 2R.$$

12. **Ans (A,C,D)**

Consider the X and Y coordinate axes along AB

and BC respectively.

Time taken to collide BC is given by

$$0 = \frac{ev\sqrt{3}}{2} - \frac{g\sqrt{3}}{2} t$$

$$\Rightarrow t = \frac{ev}{g}$$

In x-direction

$$2h = \left(\frac{v}{2} \right) t + \frac{1}{2} \left(\frac{g}{2} \right) t^2$$

$$\text{or } e^2 + 2e - \frac{8gh}{v^2} = 0$$

$$\therefore e = \frac{1}{\sqrt{3}}$$

$$(B) t = \frac{eV}{g} = \sqrt{\frac{8}{g}} s$$

$$(C) v_x = \frac{v}{2} + \frac{g}{2} t = \frac{v}{2} (1 + e)$$

$$= \sqrt{6g} \left(1 + \frac{1}{\sqrt{3}} \right) \text{ m/s}$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (8.00)**

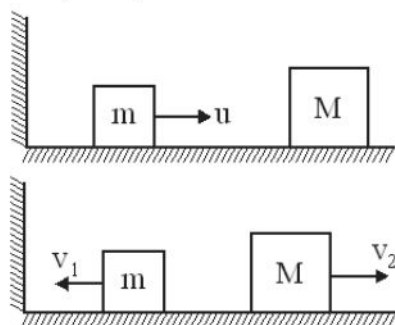
Liquid will be in pure translation

$$P_B - P_A = 2R\rho g,$$

$$K = \frac{1}{2} m V_0^2 + \frac{1}{2} \left(mR^2 + \frac{2}{3} mR^2 \right) \left(\frac{V_0}{R} \right)^2$$

$$\therefore \frac{K}{P_B - P_A} = \frac{8\pi R^2 V_0^2}{9g} \therefore n = 8$$

2. Ans (2.00)

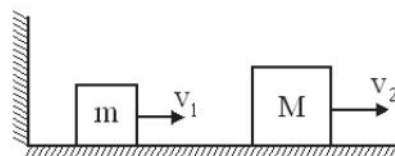


$$V_{cm} = \frac{mu}{m+M}$$

$$V_1 = -\frac{2mu}{m+M} + u$$

$$V_2 = \frac{2mu}{m+M}$$

For no collision, $V_2 \geq V_1$



$$\Rightarrow \frac{2mu}{m+M} \geq u - \frac{2mu}{m+M}$$

$$\Rightarrow \frac{4m}{m+M} \geq 1$$

$$\Rightarrow \boxed{3m \geq M}$$

3. Ans (5.00)

$$f = f_0 \frac{z^2}{n^3} \Rightarrow \frac{z^2}{n^3} = 2 \dots\dots\dots(1)$$

$$E = E_0 \frac{z^2}{n^2} \Rightarrow \frac{z^2}{n^2} = 4 \dots\dots\dots(2)$$

Solving we get $n = 2, z = 4$

$$L = mvr = \frac{nh}{2\pi}$$

$$\therefore \Delta L = \Delta n \frac{h}{2\pi}$$

Also, $\Delta L = \tau \Delta t$

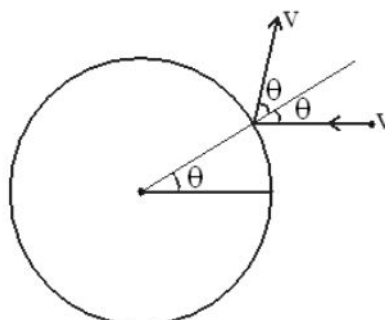
$$\therefore \tau \Delta t = \Delta n \cdot \frac{h}{2\pi} \Rightarrow \tau = \frac{\Delta n}{\Delta t} \frac{h}{2\pi}$$

$$\Rightarrow \tau = \frac{1}{7 \times 10^{-9}} \times \frac{2.1 \times 10^{-34}}{2} = 15 \times 10^{-27}$$

$$\therefore \frac{\tau}{3} \times 10^{27} = \frac{15 \times 10^{-27}}{3} \times 10^{27} = 5$$

4. Ans (8.00)

In the frame of the cylinder the particles come in with speed v and then bounce off with same speed as shown.



Force

$$F = \int_{-\pi/2}^{\pi/2} (2mv \cos \theta) \cos \theta \cdot (R d\theta \cos \theta) \ell v n$$

$$\Rightarrow \frac{F}{\ell} = \frac{8}{3} mv^2 n R \quad \therefore K = 8$$

5. Ans (2.00)

$$\vec{v} = \frac{v_0}{2} (\sqrt{3}\hat{i} + \hat{j})$$

For crossing origin,

$$K \frac{2\pi m}{qB} = \frac{2(v_0/2)}{(qE/m)}$$

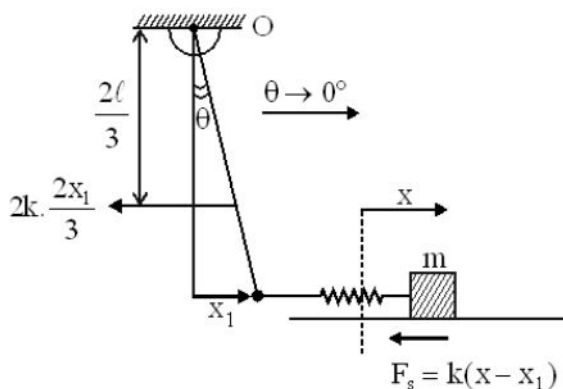
Where K has to be integer.

$$\Rightarrow B = \left(\frac{2\pi E}{v_0} \right) K$$

For minimum value $K = 1$

$$\Rightarrow B = \frac{2\pi E}{v_0}$$

6. Ans (17.00)



Light rod $\tau_0 = 0$

$$2K \cdot \frac{2x_1}{3} \cdot \frac{2\ell}{3} = K(x - x_1) \cdot \ell$$

$$\Rightarrow \frac{17}{9}x_1 = x$$

Now for the block

$$K(x - x_1) = ma$$

$$\Rightarrow K \cdot \left(x - \frac{9x}{17} \right) = ma$$

$$\Rightarrow -\frac{8K}{17m} \vec{x} = \vec{a}$$

$$\text{Time period is } T = 2\pi\sqrt{\frac{17m}{8K}}$$

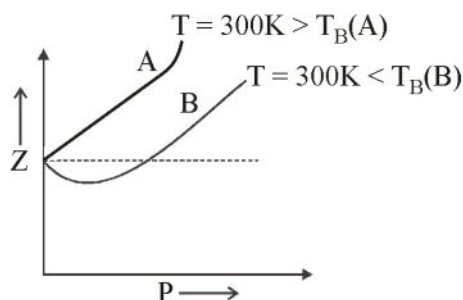
PART-2 : CHEMISTRY

SECTION-I(i)

1. Ans (C)

C.N, density and distance between closest octahedral voids ($2r$) is same for both hcp and fcc unit cells.

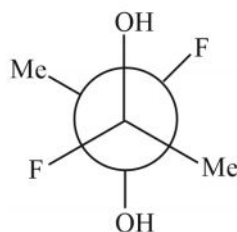
2. Ans (C)



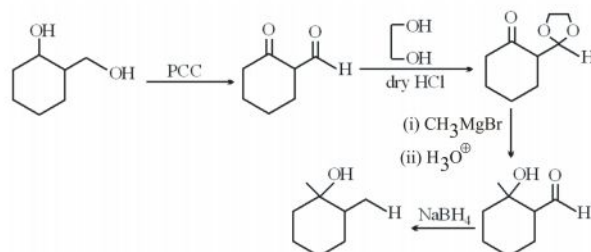
3. Ans (A)

I, II & III will have chiral gauche conformer as the most stable conformer.

IV- For the meso compound the most stable conformer is



4. Ans (B)



5. Ans (C)

Theory Based

6. Ans (A)

Theory Based.

PART-2 : CHEMISTRY

SECTION-I(ii)

7. Ans (A,B,C)

$$(a) \Delta U = W$$

$$\Rightarrow \frac{P_2 V_2 - P_1 V_1}{\gamma - 1} = -P_{\text{ext}}(V_2 - V_1)$$

$$\Rightarrow \frac{P_2 V_2 - 1 \times 20}{\frac{5}{3} - 1} = -P_2(V_2 - 20)$$

$$\Rightarrow 3(P_2 V_2 - 20) = 2(-P_2 V_2 + 20P_2)$$

$$\Rightarrow 5P_2 V_2 - 40P_2 = 60$$

$$\Rightarrow P_2 V_2 - 8P_2 = 12$$

$$\Rightarrow V_2 = \frac{12}{P_2} + 8$$

As $P_2 \rightarrow \infty$, $V_2 \rightarrow 8$ (lowest possible value)

$$(b) V_2 = \frac{12}{3} + 8 = 12L$$

$$\Delta S = nC_v \ln \left(\frac{P_2}{P_1} \right) + nC_p \ln \left(\frac{V_2}{V_1} \right)$$

$$= 1 \times \frac{3R}{2} \ln \left(\frac{3}{1} \right) + 1 \times \frac{5R}{2} \ln \left(\frac{12}{20} \right)$$

$$= \left[3 \ln 3 + 5 \ln \left(\frac{3}{5} \right) \right] \text{ Cal K}^{-1}$$

$$= 0.8 \text{ Cal K}^{-1}$$

(c) For adiabatic process,

$$\Delta S_{\text{Surr}} = 0$$

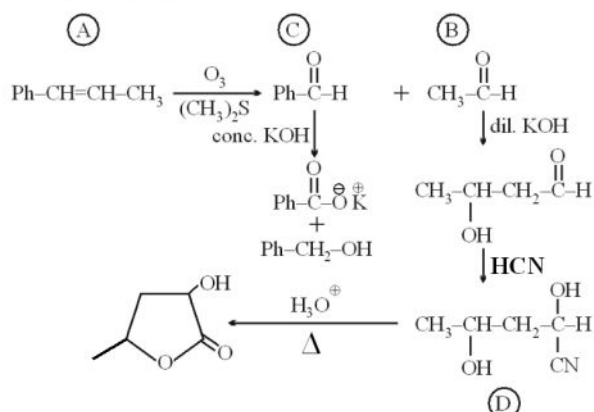
$$\Delta S_{\text{sys}} = +ve$$

(d) $TV^{\gamma-1}$ or $TV^{(R/CV)}$ is only valid for reversible processes.

8. Ans (B,D)

HNO_3 and H_2O form maximum boiling azeotrope and upon distillation, the distillate moves farther away from azeotrope composition, i.e. $y < \chi$ for $\chi < \chi_{\text{azeotrope}}$

9. Ans (B,C,D)



11. Ans (A,D)

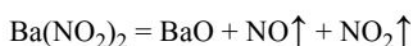
(A) Only Cl is ionisable.

(B) Fe^{+3} has odd number of electrons ($3d^5$), so paramagnetic.

(C) Only shows optical isomerism.

(D) As C_2H_4 bonds to Pt it's an organometallic compound.

12. Ans (A,B,C)

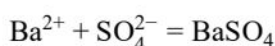


(A) (D)

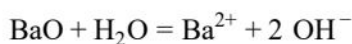
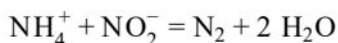
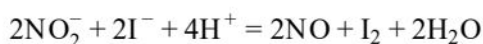


(B)

NH_4HSO_4 boils without further decomposition at 490°C .



(C)



(D)



(E)

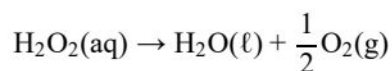


(E) (D)

PART-2 : CHEMISTRY

SECTION-II

1. **Ans (3.00)**



$$\Delta H_r = \Delta H_{f_{\text{H}_2\text{O}}}^\circ - \Delta H_{f_{\text{H}_2\text{O}_2}}^\circ = -287 - (-187)$$

$$= -100 \text{ kJ/mol.}$$

$$\Rightarrow -\frac{d[\text{H}_2\text{O}_2]}{dt} = k[\text{H}_2\text{O}_2]$$

$$= 7.5 \times 10^{-4} \times 0.02 \text{ mol L}^{-1} \text{ s}^{-1}.$$

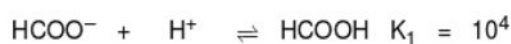
∴ Heat evolved per second

$$= (2 \times 7.5 \times 10^{-4} \times 0.02 \times 100 \times 10^3) \text{ Js}^{-1} = 3 \text{ Js}^{-1}.$$

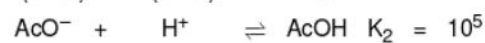
2. **Ans (4.50)**

Initial concentration of $[\text{HCOO}^-]$, $[\text{CH}_3\text{COO}^-]$

and $[\text{H}^+] = 1\text{M}$



$$(1-x) \quad (1-x) \quad x$$



$$(1-y) \quad (1-x-y) \quad y$$

∴ The equilibrium constants are very large, H^+

acts as the limiting reagent, and $1-x-y \approx 0$

$$\Rightarrow x+y \approx 1 \dots(\text{i})$$

$$K_1 = 10^4 = \frac{x}{(1-x)[\text{H}^+]}$$

$$\Rightarrow \frac{x}{1-x} = 10^4[\text{H}^+] \dots(\text{ii})$$

$$K_2 = 10^5 = \frac{y}{(1-y)[\text{H}^+]}$$

$$\Rightarrow \frac{1-y}{y} = \frac{1}{10^5[\text{H}^+]}$$

$$\text{Also } y = 1-x$$

$$\Rightarrow \frac{x}{1-x} = \frac{1}{10^5[\text{H}^+]} \dots(\text{iii})$$

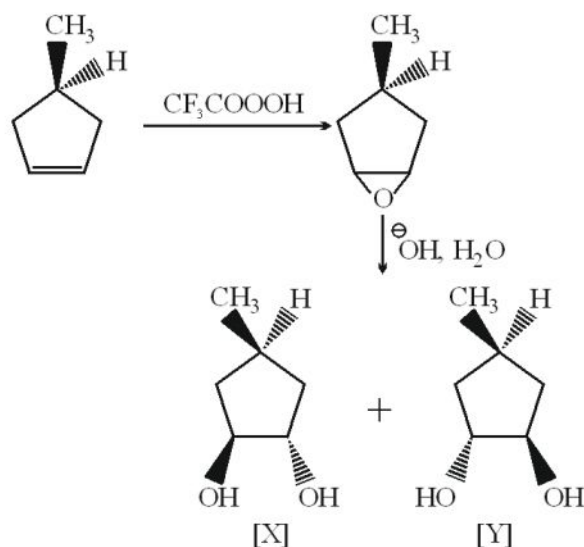
$$(\text{ii}) = (\text{iii}) \Rightarrow 10^4[\text{H}^+] = \frac{1}{10^5[\text{H}^+]}$$

$$\Rightarrow [\text{H}^+]^2 = 10^{-9}$$

$$\Rightarrow \text{pH} = 4.5$$

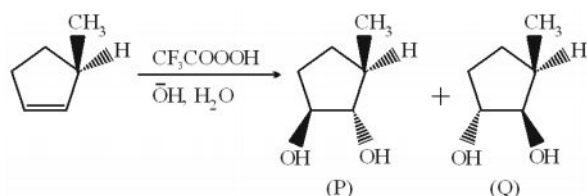
3. **Ans (3.00)**

(i)



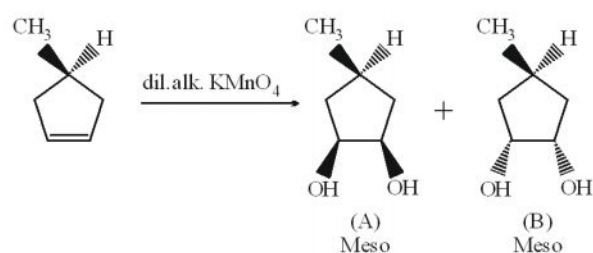
X and Y are enantiomer

(ii)



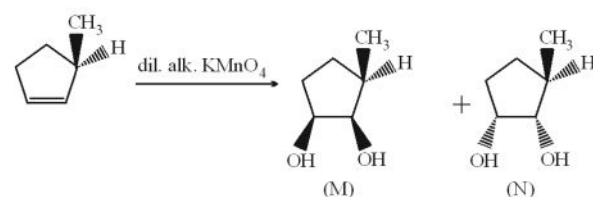
P and Q are diastereomer

(iii)



A and B are diastereomer

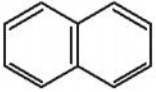
(iv)

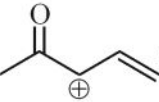


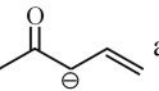
Correct statements are (1), (3) & (6)

4. Ans (6.00)

(1), (3) extended conjugation stabilizes more than cross conjugation

(2)  Both rings are aromatic

(4) In , cation is destabilized by >C=O group.

(5) In , anion is stabilized by >C=O group.

(6) Cation is not delocalised.

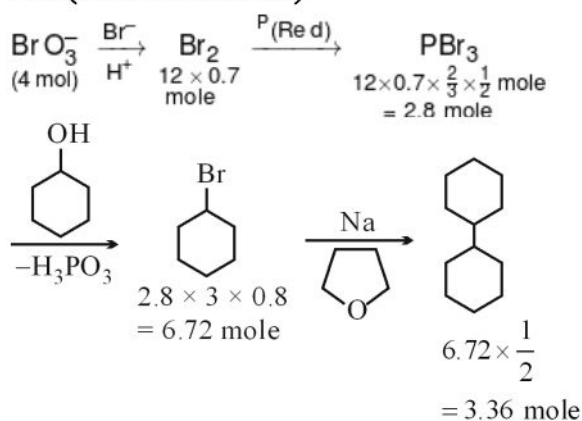
(7) 3° carbocation is more stable than 2° carbocation

(8) Uncharged resonating structure is more stable than charged.

(9) Cation is de-stabilised by -I effect.

Correct statements are 2,3,4,6,7, & 8.

5. Ans (556.10 to 557.80)



3.36 mole of D = $3.36 \times 166 = 557.76$

6. Ans (8.90 to 9.10)

dipole moment of HCHO = $0.4 + 2.3 = 2.7$ D

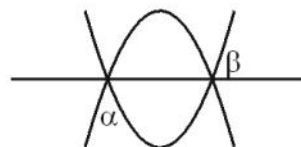
$2.7 \text{ D} = 2.7 \times 3.335 \times 10^{-30} \text{ C m} = 9.00 \times 10^{-30} \text{ C m}$

PART-3 : MATHEMATICS

SECTION-I(i)

1. Ans (D)

$$(x^2 + 5x - b^2)(-2x^2 + ax + 6) = 0$$



$$\Rightarrow \text{both roots are in common } \frac{1}{-2} = \frac{5}{a} = \frac{-b^2}{6}$$

$$\Rightarrow a = -10, b^2 = 3$$

$$\text{Eqn. } x^2 + 5x - 3 = 0$$

$$\Rightarrow a + b = -5, \alpha\beta = -3$$

2. Ans (C)

Range of $f(x)$ contains $[0, \infty)$

$\therefore$ Range of $\text{gof}(x)$ is $(0, \pi]$

$\Rightarrow$ Number of integers are 3.

3. Ans (D)

$$\lim_{x \rightarrow 0^+} \frac{((1 + \tan x)^{\cot x} + e^{\cos x})((1 + \tan x)^{\cot x} - e^{\cos x})}{\frac{1 - \cos(\ln(1 + \sin \sqrt{x}))}{\ln^2(1 + \sin \sqrt{x})} \times \frac{\ln^2(1 + \sin \sqrt{x})}{(\sin \sqrt{x})^2} \times \frac{\sin^2 \sqrt{x}}{(\sqrt{x})^2} \times x}$$

$$4e \lim_{x \rightarrow 0^+} \frac{(1 + \tan x)^{\cot x} - e^{\cos x}}{x}$$

$$\Rightarrow 4e \lim_{x \rightarrow 0^+} \frac{e^{1 - \frac{1}{2}\tan x + \frac{1}{3}\tan^2 x + \dots} - e^{\cos x}}{x}$$

$$= 4e \lim_{x \rightarrow 0^+} \frac{e^{\cos x} \left[e^{(1 - \frac{1}{2}\tan x + \frac{1}{3}\tan^2 x + \dots) - \cos x} - 1 \right]}{x}$$

$$= 4e^2 \lim_{x \rightarrow 0^+} \frac{\left(1 - \frac{1}{2}\tan x + \frac{1}{3}\tan^2 x + \dots \right) - \cos x}{x}$$

$$= 4e^2 \left(-\frac{1}{2} \right) = -2e^2$$

4. Ans (A)

$$\begin{aligned} & \int \left(e^x \left(\frac{\sin x + \cos x}{(\sin x + \cos x)^2} - \frac{\cos x - \sin x}{(\sin x + \cos x)^2} \right) \right. \\ & \left. + e^x \left(\frac{\sin x - \cos x}{(\sin x - \cos x)^2} - \frac{\cos x + \sin x}{(\sin x - \cos x)^2} \right) \right) dx \\ &= e^x \left[\frac{\sin x + \cos x}{(\sin x + \cos x)^2} \right] + e^x \left[\frac{\sin x - \cos x}{(\sin x - \cos x)^2} \right] + C \\ &= e^x \left[\frac{1}{\sin x + \cos x} + \frac{1}{\sin x - \cos x} \right] \\ &= e^x \left[\frac{2 \sin x}{\sin^2 x - \cos^2 x} \right] + C \end{aligned}$$

5. Ans (B)

$$S = (\sin^3 \theta + \sin^3 3\theta + \dots + \sin^3 19\theta) +$$

$$(\cos^3 \theta + \cos^3 3\theta + \dots + \cos^3 19\theta)$$

$$\text{Using, } \sin^3 \theta = \frac{3 \sin \theta - \sin 3\theta}{4} \text{ and}$$

$$\cos^3 \theta = \frac{\cos 3\theta + 3 \cos \theta}{4}$$

$$\text{Let } S_1 = (\sin^3 \theta + \sin^3 3\theta + \dots + \sin^3 19\theta)$$

$$= \frac{3}{4} (\sin \theta + \sin 3\theta + \dots + \sin 19\theta)$$

$$- \frac{1}{4} (\sin 3\theta + \sin 9\theta + \dots + \sin 57\theta)$$

$$= \frac{3}{4} \left(\frac{\sin 10\theta}{\sin \theta} \cdot \sin 10\theta \right)$$

$$- \frac{1}{4} \left(\frac{\sin 30\theta}{\sin 3\theta} \cdot \sin 30\theta \right)$$

$$\text{For } \theta = 15, S_1 = \frac{3}{4} \left(\frac{1}{4} \cdot \frac{2\sqrt{2}}{\sqrt{3}-1} \right)$$

$$- \frac{1}{4} (\sqrt{2}) = \frac{3(\sqrt{3}+1)-4}{8\sqrt{2}}$$

$$\text{and let } S_2 = (\cos^3 \theta + \cos^3 3\theta + \dots +$$

$$\cos^3 19\theta)$$

$$= \frac{3}{4} \left(\frac{\sin 10\theta}{\sin \theta} \cdot \cos 10\theta \right) + \frac{1}{4} \left(\frac{\sin 30\theta \cos 30\theta}{\sin 3\theta} \right)$$

$$\text{For } \theta = 15,$$

$$S_2 = \frac{3}{4} \left(\frac{-\sqrt{3}}{4} \cdot \frac{2\sqrt{2}}{\sqrt{3}-1} \right) + \frac{1}{4} (0)$$

$$= \frac{-3\sqrt{3}(\sqrt{3}+1)}{8\sqrt{2}}$$

$$S = S_1 + S_2 = \frac{3(\sqrt{3}+1)-4}{8\sqrt{2}} - \frac{3\sqrt{3}(\sqrt{3}+1)}{8\sqrt{2}}$$

$$= \frac{3\sqrt{3}-1-9-3\sqrt{3}}{8\sqrt{2}} = \frac{-10}{8\sqrt{2}} = \frac{-5}{\sqrt{32}} \equiv \frac{-a}{\sqrt{b}}$$

$$\therefore a = 5, b = 32$$

$$\Rightarrow |b - 6a| = 2.$$

6. Ans (B)

$$\therefore f(x) \text{ is non-decreasing}$$

$$\Rightarrow f'(x) \geq 0$$

$$\Rightarrow 3x^2 + 2ax + b \geq 0 \Rightarrow D \leq 0$$

$$\Rightarrow 4a^2 - 4.3.b \leq 0 \Rightarrow a^2 \leq 3b$$

$$a = 1, b = 1, 2, 3, 4, 5, 6$$

$$a = 2, b = 2, 3, 4, 5, 6$$

$$a = 3, b = 3, 4, 5, 6$$

$$a = 4, b = 6$$

$$\Rightarrow a \text{ \& b can vary in 16 ways}$$

$$\Rightarrow a, b, c \text{ can vary in } 16 \times 6 = 96 \text{ ways}$$

$$\therefore f(x) \text{ has all roots real, } f(x) \text{ must be of the form } (x + \alpha)^3$$

$$\text{only possible case } (x+1)^3 \Rightarrow a = 3, b = 3, c = 1$$

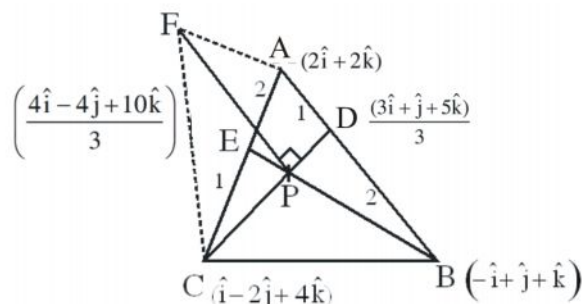
$$\Rightarrow \text{required probability} = \frac{1}{96}$$

PART-3 : MATHEMATICS

SECTION-I(ii)

7. Ans (B,C,D)

$$CD; \vec{r} = (\hat{i} - 2\hat{j} + 4\hat{k}) + \lambda (7\hat{j} - 7\hat{k})$$



$$BE; \vec{r} = (-\hat{i} + \hat{j} + \hat{k}) + m (7\hat{i} - 7\hat{j} + 7\hat{k})$$

∴ Point of intersection of CD & BE is P

$$\therefore P = (\hat{i} - \hat{j} + 3\hat{k})$$

$$\vec{AB} \times \vec{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -3 & 1 & -1 \\ -1 & -2 & 2 \end{vmatrix}$$

$$= \hat{i}(0) - \hat{j}(-7) + \hat{k}(7) = 7\hat{j} + 7\hat{k}$$

$$\vec{PB} \times \vec{PC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -2 & 2 & -2 \\ 0 & 1 & -1 \end{vmatrix}$$

$$= (-2)\hat{j} + (-2)\hat{k} = -2\hat{j} - 2\hat{k}$$

Area of tetrahedral ABCF

$$= \frac{1}{3}(\text{Base Area}) \times \text{Height} = \frac{1}{3} \left(\frac{7\sqrt{2}}{2} \right) \times \sqrt{2} = \frac{7}{3}$$

$$\vec{PF} = \pm \sqrt{2} \left(\frac{\hat{j} + \hat{k}}{\sqrt{2}} \right) = (\hat{j} + \hat{k}) = \vec{F} - \vec{P}$$

$$\therefore \vec{F} = (\hat{i} - \hat{j} + 3\hat{k}) \pm (\hat{j} + \hat{k}) = \hat{i} + 4\hat{k}; \hat{i} - 2\hat{j} + 2\hat{k}$$

$$AF; \vec{r} = 2(\hat{i} + \hat{k}) + \lambda(\hat{i} - 2\hat{k}); \vec{r} = 2(\hat{i} + \hat{k}) + \lambda(\hat{i} + 2\hat{j})$$

8. Ans (A,C,D)

Any point on the line $x + y = 1$ will be $(t, 1 - t)$.

Equation of chord of the ellipse with this point

as a midpoint

$$\frac{tx}{2a^2} + \frac{y(1-t)}{2} = \frac{t^2}{2a^2} + \frac{(1-t)^2}{2}$$

it passes through $(a, -1)$ so that

$$\frac{at}{2a^2} + \frac{(t-1)}{2} = \frac{t^2}{2a^2} + \frac{(1-t)^2}{2}$$

$$at + a^2(t-1) = t^2 + a^2(1-t)^2$$

$$(a^2 + 1)t^2 - (3a^2 + a)t + 2a^2 = 0$$

Since we want two such chords

$$D > 0$$

$$(3a^2 + a)^2 - 4(a^2 + 1) \cdot 2a^2 > 0$$

$$(3a + 1)^2 - 8(a^2 + 1) > 0$$

$$a^2 + 6a - 7 > 0$$

$$(a + 7)(a - 1) > 0$$

$$a \in (-\infty, -7) \cup (1, \infty).$$

9. Ans (A,B,D)

$$f(x+y) = f(x) + f(y) + (x+y) \cdot e^{x+y} - x \cdot e^x - y \cdot e^y + 4xy$$

$$\text{Put } x = y = 0 \Rightarrow f(0) = 0$$

Differentiate partially w.r.t. x

$$f(x+y) = f(x) + (x+y) \cdot e^{x+y} + e^{x+y} - x \cdot e^x - e^x + 4y$$

$$\text{Put } y \rightarrow -x$$

$$f(0) = f(x) + 1 - x \cdot e^x - e^x - 4x$$

$$f(x) = x \cdot e^x + e^x + 4x$$

$$f(x) = x \cdot e^x + 2x^2$$

10. Ans (C,D)

$$\int_0^{\pi} [\{x\} + [\cot x]] dx = \int_0^{\pi} ([\{x\}] + [\cot x]) dx$$

$$= \int_0^{\pi} [\cot x] dx$$

$$\text{Now, } I = \int_0^{\pi} [\cot x] dx = \int_0^{\pi} [\cot (\pi - x)] dx$$

$$2I = \int_0^{\pi} ([\cot x] + [-\cot x]) dx$$

$$2I = \int_0^{\pi} (-1) dx = -\pi \Rightarrow I = -\frac{\pi}{2}$$

11. Ans (A,B,C)

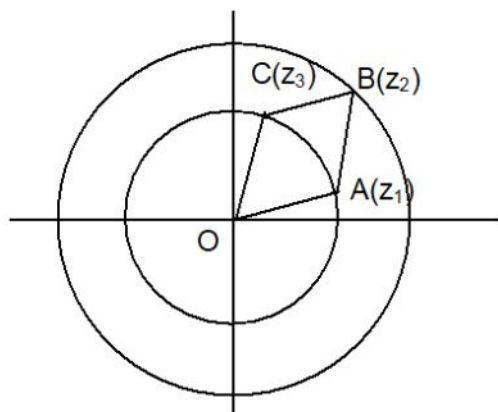
$$\text{Given that } \arg(z_2 - z_1) = \frac{\pi}{3}$$

$$\text{since } \arg z_3 = \arg(z_2 - z_1) = \frac{\pi}{3}$$

$$\Rightarrow z_3 = |z_3| e^{i\pi/3}$$

$$\Rightarrow z_3 = 2 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)$$

$$\Rightarrow z_3 = 2 \left(\frac{1}{2} + \frac{\sqrt{3}i}{2} \right) = (1 + \sqrt{3}i)$$



$$\text{The mid point of diagonals} = \frac{z_1 + z_3}{2} = \frac{z_2 + 0}{2}$$

$$\Rightarrow z_1 + z_3 = z_2 \quad \dots (1)$$

since $z_3 - z_1$ is perpendicular to z_2

$$\Rightarrow \frac{z_3 - z_1}{z_2} = ki \Rightarrow z_3 - z_1$$

$$= ki z_2 \quad (k \in \mathbb{R}) \quad \dots (2)$$

$$\text{From (1) and (2) } z_1 = \left(\frac{1 - ki}{1 + ki} \right) z_3$$

$$\text{and } z_2 = \frac{2z_3}{1 + ki}$$

$$\Rightarrow |z_2| = \frac{2|z_3|}{|1 + ki|} \Rightarrow 3 = \frac{2 \times 2}{\sqrt{1 + k^2}}$$

$$\Rightarrow 1 + k^2 = \frac{16}{9}$$

$$\Rightarrow k = \pm \frac{\sqrt{7}}{3} \Rightarrow z_2 = \left(\frac{2}{1 \pm \frac{\sqrt{7}}{3}i} \right) \cdot (1 + \sqrt{3}i)$$

$$\Rightarrow z_2 = \frac{3(2 + 2\sqrt{3}i)}{3 + \sqrt{7}i} \text{ or } \frac{3(2 + 2\sqrt{3}i)}{3 - \sqrt{7}i} \text{ and}$$

$$z_1 = \left(\frac{3 - \sqrt{7}i}{3 + \sqrt{7}i} \right) (1 + \sqrt{3}i)$$

$$\text{or } \left(\frac{3 + \sqrt{7}i}{3 - \sqrt{7}i} \right) (1 + \sqrt{3}i)$$

As two such rhombuses are possible, hence one

set of values is

$$z_1 = \left(\frac{3 - \sqrt{7}i}{3 + \sqrt{7}i} \right) (1 + \sqrt{3}i)$$

$$z_2 = \frac{6 + 6\sqrt{3}i}{3 + \sqrt{7}i} \text{ and } z_3 = (1 + \sqrt{3}i) \text{ and another}$$

set of the values is

$$z_1 = \left(\frac{3 + \sqrt{7}i}{3 - \sqrt{7}i} \right) (1 + \sqrt{3}i),$$

$$z_2 = \left(\frac{6 + 6\sqrt{3}i}{3 - \sqrt{7}i} \right) \text{ and } z_3 = (1 + \sqrt{3}i).$$

12. Ans (B,C,D)

$$f'(x) = -3x^2 + 2px + q > 0 \quad \forall x \in (-2, 1)$$

$$\text{Sum of roots} \Rightarrow \frac{2p}{3} = -1 \Rightarrow p = -\frac{3}{2}$$

$$\text{and product of roots} \Rightarrow \frac{-q}{3} = -2 \Rightarrow q = 6$$

$$\therefore (p + q) = \frac{9}{2}.$$

PART-3 : MATHEMATICS

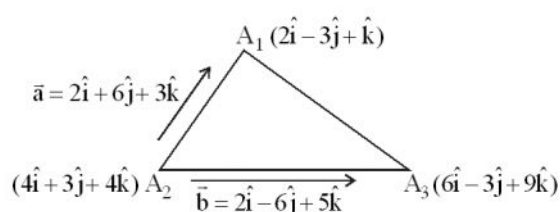
SECTION-II

1. Ans (9.50)

$$\text{Normal vector of plane } \vec{n}_1 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 3 & 2 \\ 2 & 3 & -1 \end{vmatrix}$$

$$= -9\hat{i} + 3\hat{j} - 9\hat{k}$$

$$\vec{a} \times \vec{b} = \vec{n}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 6 & 3 \\ 2 & -6 & 5 \end{vmatrix} = 48\hat{i} - 4\hat{j} - 24\hat{k}$$



$$\text{Required area} = \frac{1}{2} |\vec{a} \times \vec{b}| \hat{n}_2 \cdot \hat{n}_1$$

$$= \frac{1}{2} 4\sqrt{181} \cdot \frac{228}{4\sqrt{181}\sqrt{171}} = 2\sqrt{19} \equiv a\sqrt{b}$$

$$\therefore \left(\frac{b}{a}\right) = 9.50.$$

2. Ans (119.00)

Let page numbers $k, k+1, k+2, k+3$ are torn off.

$$\Rightarrow \frac{n(n+1)}{2} - (4k+6) = 1596$$

$$\Rightarrow n^2 + n - 4(2k+3) = 3192$$

$$\Rightarrow n^2 + n - 3192 = 4(2k+3)$$

$$\Rightarrow (n+57)(n-56) = 4(2k+3)$$

$$\text{if } n = 59, k = 42$$

$$\text{if } n = 60, k = 57$$

$$\text{Sum} = 59 + 60 = 119$$

3. Ans (0.50)

Let $A \equiv (\alpha, \beta)$ the equation of the normal to the parabola at $(at^2, 2at)$ is

$$y + tx = 2at + at^3$$

which passes through (α, β)

$$\text{So, } at^3 + (2a - \alpha)t - \beta = 0$$

$$\Rightarrow t_1 + t_2 + t_3 = 0 \quad \dots (1)$$

So, we have

$$P(at_1^2, 2at_1), Q(at_2^2, 2at_2) \text{ and } R \equiv (at_3^2, 2at_3)$$

$$C_1 \equiv \text{Circle drawn on AP as diameter} = (x - \alpha)$$

$$(x - at_1^2) + (y - \beta)(y - 2at_1) = 0.$$

$$C_2 \equiv \text{Circle drawn on AQ as diameter} = (x - \alpha)$$

$$(x - at_2^2) + (y - \beta)(y - 2at_2) = 0.$$

$$\text{Common chord} \equiv (t_1 + t_2)x + 2y - \alpha(t_1 + t_2) - 2\beta = 0 \quad (\text{as } t_1 \neq t_2)$$

$$\therefore m_1 = -\left(\frac{t_1 + t_2}{2}\right) = \frac{t_3}{2} \quad (\text{from (1)})$$

Tangent to the parabola at R

$$t_3y = x + at_3^2 \Rightarrow m_2 = \frac{1}{t_3}$$

$$\text{so, } m_1 \times m_2 = \frac{1}{2}.$$

4. Ans (1206.00)

$$\text{Case-1: } 4, 1, 0, 0, 0, 0, 0, 0 \rightarrow \frac{9!}{7!} = 72$$

$$\text{Case-2: } 2, 2, 2, 2, 1, 0, 0, 0 \rightarrow \frac{9!}{4! \cdot 4!} = 630$$

$$\text{Case-3: } 2, 2, 2, 1, 1, 1, 1, 0 \rightarrow \frac{9!}{3! \cdot 5!} = 504$$

$$\therefore \text{Number of matrices in } S = 72 + 630 + 504 = 1206.$$

5. Ans (19.20)

$$\int_0^1 (6x^4 \cdot f(x) - f^2(x) - 9x^8 + 9x^8) dx = 1$$

$$\int_0^1 (9x^8 - (f(x) - 3x^4)^2) dx = 1$$

$$1 - \int_0^1 (f(x) - 3x^4)^2 dx = 1$$

$$\int_0^1 (f(x) - 3x^4)^2 dx = 0$$

$$\therefore f(x) = 3x^4$$

$$\therefore \text{Required Area} = \int_0^2 (3x^4) dx = \frac{3}{5} x^5 \Big|_0^2$$

$$= \frac{3}{5} \times (32 - 0) = 19.20.$$

6. Ans (5.00)

$$u = \sin^7 x, v = \cos x$$

$$\frac{du}{dv} = \frac{7\sin^6 x \cos x}{-\sin x} = -7\sin^5 x \cos x$$

$$\frac{d^2u}{dv^2} = 7[5\sin^3 x \cos^2 x - \sin^5 x]$$

$$\frac{d^3u}{dv^3} = -\frac{105}{4} \sin 4x$$

$$\text{Let } \frac{d^3u}{dv^3} = z \Rightarrow z = \frac{-105}{4} \sin 4x$$

$$\Rightarrow \frac{dz}{dx} = \frac{-105}{4} (4 \cos 4x)$$

$$\Rightarrow \frac{d^2z}{dx^2} = -105 (-4 \sin 4x)$$

$$\frac{d^2z}{dx^2} \Big|_{x=\frac{\pi}{8}} = 105 \times 4$$

$$= 2^2 \cdot 3^1 \cdot 5^1 \cdot 7^1 \equiv 2^{\lambda_1} \cdot 3^{\lambda_2} \cdot 5^{\lambda_3} \cdot 7^{\lambda_4}$$

$$\Rightarrow \lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 = 5.$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE (Advanced)

FULL SYLLABUS

08-05-2024

JEE(Main + Advanced) : ENTHUSIAST COURSE ALL STAR BATCH (SCORE ADVANCED)

ANSWER KEY

PAPER-2 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	A	B	C	C		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,C	A,C,D	A,B	A,C	A,D	B,C,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	0.35	4.83	333.33	316.67	100.69	103.04
SECTION-II (ii)	Q.	7	8	9			
	A.	5	1	7			

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4		
	A.	C	D	D	C		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,B,C	A,B,C	C	A,B,D	A,C	B,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	132.00	116.00	4.00	3.00	9.74	2.77
SECTION-II (ii)	Q.	7	8	9			
	A.	7	6	5			

PART-3 : MATHEMATICS

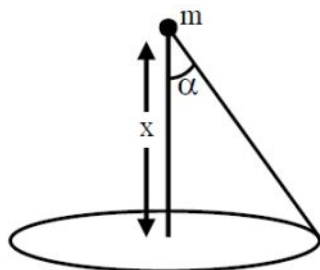
SECTION-I (i)	Q.	1	2	3	4		
	A.	B	A	C	B		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,B,D	A,B,C	A,D	B,C,D	A,B,C	A,B,C,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	25.00	1.57	0.86	0.34	3.73	0.45
SECTION-II (ii)	Q.	7	8	9			
	A.	2	8	7			

HINT – SHEET

PART-1 : PHYSICS

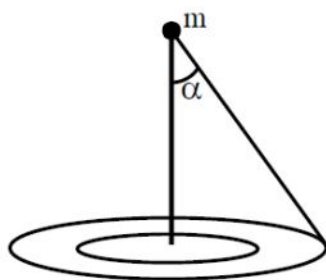
SECTION-I (i)

1. Ans (A)



$$g = 2\pi G\sigma [1 - \cos \alpha]$$

$$g = 2\pi G\sigma \left[1 - \frac{x}{\sqrt{R^2 + x^2}} \right] = 2\pi G \left[1 - \frac{x}{R} \right]$$

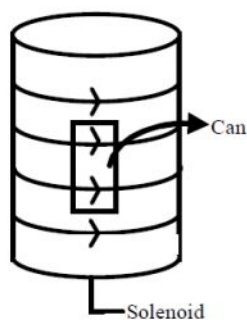


$$g = 2\pi G\sigma \left[\left(1 - \frac{x}{R} \right) - \left(1 - \frac{x}{R/2} \right) \right]$$

$$g = \frac{2\pi G\sigma}{R} x$$

$$T = \sqrt{\frac{2\pi R}{G\sigma}}$$

2. Ans (B)

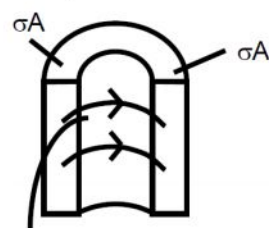


Emf is induced in the can

$$\text{Emf} = \frac{dB}{dt} \pi r^2 = K \pi r^2$$

$$R = \frac{\rho \times 2\pi r}{A}$$

$$i = \frac{r}{2\rho} AK$$



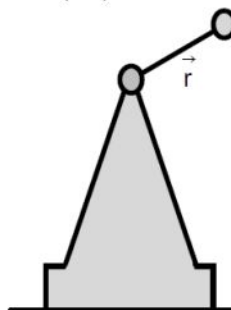
$$i2rB$$

From the diagram

$$i \times 2rB = 2\sigma A$$

$$t = \frac{2\sigma\rho}{r^2 K^2}$$

3. Ans (C)



$$\vec{F} \text{ on ball} + m\vec{g} = -m\omega^2 \vec{r}$$

$$\vec{F} \text{ on platform} = m\vec{g} + m\omega^2 \vec{r}$$

$$m\omega^2 r = \frac{\mu 5mg}{\sqrt{1 + \mu^2}} \omega_0 = 5 \text{ rad/s}$$

4. Ans (C)

For perfect conductor $R = 0$ $E_{\text{net}} = 0$ in loop

$$\phi_{\text{net}} = \text{constant}$$

$$\phi_{\text{net}} = \phi_{\text{self}} + \phi_k + \phi_{\text{ext}}$$

$$= L_i + L_{ki} + \vec{B} \cdot \vec{A} = L_{\text{net}i} + \vec{B} \cdot \vec{A}$$

$$L_K = \frac{m l}{n e^2 A} \quad (\text{Where } m = \text{mass of one electron, } l$$

$$= \text{loop length})$$

$$\text{As } \phi \text{ is constant } \phi_i = \phi_f$$

$$\text{Initially } i = 0 \text{ \& } \vec{B} \perp \vec{A}$$

$$\phi_{\text{net}} = 0$$

$$\text{Finally } \phi = -BA' + L_{\text{net}}i$$

$$i = \frac{B\pi r^2}{L_{\text{net}}} = \frac{B\pi r^2}{L + \frac{m \times 2\pi r}{n e^2 A}}$$

PART-1 : PHYSICS

SECTION-I (ii)

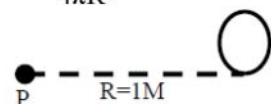
5. Ans (A,C)

$$L_{\text{eq}} = \frac{L_1 L_2 - M^2}{L_1 + L_2 - 2M} \quad \&$$

$$L_{\text{eq}} = \frac{L_1 L_2 - M^2}{L_1 + L_2 - 2M} \quad (\text{Opposing})$$

6. Ans (A,C,D)

$$l = \frac{P}{4\pi R^2}$$



$$\text{Energy incident} = l\pi r^2$$

$$\text{No. of Photons incident/sec} = \frac{P \times \pi r^2}{4\pi R^2} \times \frac{\lambda}{hc}$$

$$\text{For } \lambda_1 \text{ no. of photon incident/sec } 2.5 \times 10^{15}$$

Photo electric effect for λ_1 & λ_2 only.

$$h\nu = \phi + K_{\text{max}} \Rightarrow K_{\text{max}} = 0 \quad (\text{For } \lambda_1)$$

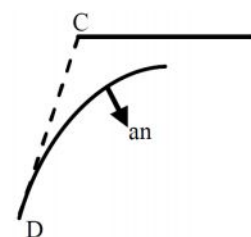
$$E_2 = \frac{12400}{4133.33} = 3\text{eV} \Rightarrow \phi = \frac{12400}{4960} = 2.5\text{eV}$$

$$V_s = 0.5 \text{ volt}$$

$$\frac{KQ}{r} = V_s \Rightarrow Q_2 = ne$$

7. Ans (A,B)

Acceleration of dog = Normal acceleration.



$$\Delta \vec{A} = \vec{A}_f - \vec{A}_i = \vec{A}_i + \Delta \vec{A}$$

$$a = \frac{dv}{dt} = \frac{v d\theta}{dt} = v\omega = \frac{v \sin \theta}{r}$$

$$R = \frac{v^2}{a_n} = \frac{vr}{u \sin \theta}$$

8. Ans (A,C)

$$\vec{F} = -2k\vec{r} \text{ toward } 0.$$

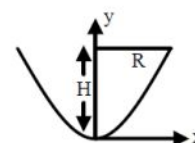
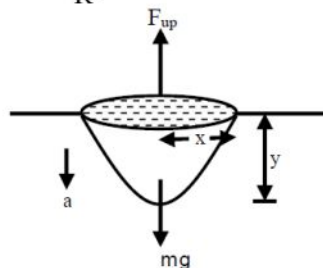
$$\vec{\tau}_0 = 0$$

$$P.t = \frac{1}{2}k \left(r^2 + \frac{r^2}{4} \right) = 1.25 Kr^2$$

9. Ans (A,D)

Let $Y = ax^2$ (A parabola)

$$Y = \frac{H}{R^2} x^2 \quad (\text{from given data})$$



$$v = \frac{dy}{dt}$$

$$mg - \rho_L \cdot \frac{1}{2} \pi x^2 y \cdot g = m \cdot \frac{dv}{dt}$$

$$\frac{dv}{dt} = g - \frac{\pi x^2 y g \rho_L}{2 \rho_s \frac{\pi R^2 H}{2}} g \Rightarrow g - \frac{\rho_L}{\rho_s} \frac{x^2}{R^2} \times \frac{yg}{H}$$

$$v \cdot \frac{dv}{dy} = g - \frac{\rho_L}{\rho_s} gy^2$$

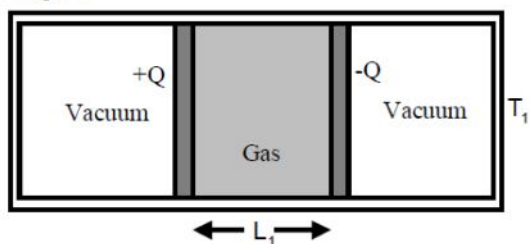
$$\int_0^v v dv = \int_0^H \left(g - \frac{\rho_L}{\rho_s} gy^2 \right) dy$$

$$0 = gH - \frac{g\rho_L}{3\rho_s} H$$

$$\frac{\rho_L}{\rho_s} = 3$$

10. Ans (B,C,D)

Stage 1

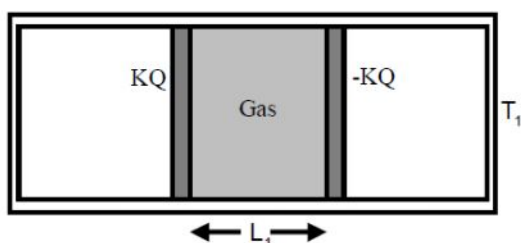


$$P_1 A L_1 = n R T_1 \quad \frac{Q_2}{2A\epsilon_0} = P_1 A = \frac{n R T_1}{L_1}$$

(Force Balancing)

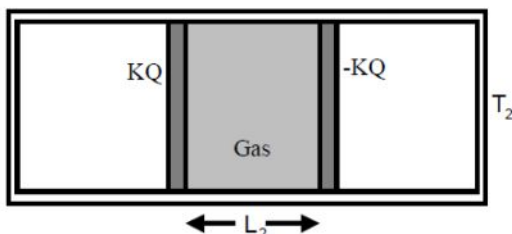
Stage 2

Charge is suddenly changed



System is not in equilibrium

Stage 3



$$L_2 C L_1$$

Finally system in equilibrium

$$P_2 A L_2 = n R T_2 \quad \frac{K^2 Q^2}{2A\epsilon_0} = P_2 A = \frac{n R T_2}{L_2}$$

Energy of the system is conserved between stage 2 and 3.

$$\Rightarrow \frac{1}{2} \epsilon_0 E_2^2 A L_1 + \frac{f}{2} n R T_1 = \frac{1}{2} \epsilon_0 E_2^2 A L_2 + \frac{f}{2} n R T_2$$

On solving all the above equation

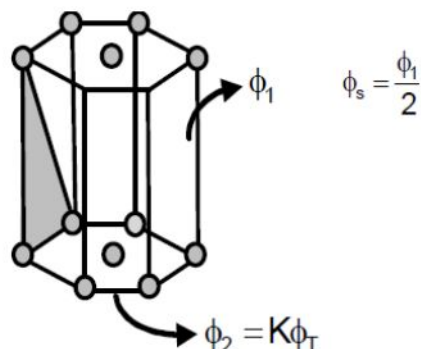
$$\frac{T_2}{T_1} = \frac{2K^2 + f}{2 + f}$$

$$w_{\text{ext}} = (K^2 - 1) \frac{Q^2 L_1}{2A\epsilon_0} = (K^2 - 1) n R T_1$$

PART-1 : PHYSICS

SECTION-II (i)

2. Ans (4.83)

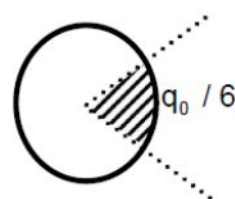


$$6\phi_1 + 2\phi_2 = \phi_{\text{total}}$$

$$6\phi_1 + 2K\phi_{\text{total}} = \phi_{\text{total}}$$

$$\phi_1 = \frac{(1 - 2K) \phi_{\text{total}}}{6} \Rightarrow \phi_{\text{total}} = \frac{q_{\text{in}}}{\epsilon_0}$$

Top view



$$q_{\text{in}} = 12 \cdot \frac{q_0}{6} + 2 \cdot \frac{q_0}{2}$$

$$\Rightarrow 3q_0 = q_{\text{in}} \Rightarrow 3d_0 \frac{4}{3} \pi R_0^3 = q_{\text{in}}$$

$$\phi_1 = \frac{(1 - 2k)}{6} \times \frac{4\pi R_0^3 d_0}{\epsilon_0}$$

$$\phi_s = \frac{(1 - 2k)}{3} \frac{\pi R_0^3 d_0}{\epsilon_0} = 0.35$$

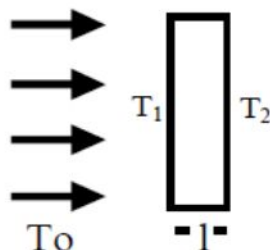
$$\text{Average charge density} = \frac{q_{\text{in}}}{V_{\text{prism}}}$$

$$d = \frac{8\pi}{3\sqrt{3}} \frac{d_0 R_0^3}{L_0^2 W_0} = 4.83$$

4. Ans (316.67)

Left face

Power from sun = P



$$\frac{d\theta}{dt} \propto (T_1 - T_0) \quad \frac{d\theta}{dt} = \lambda (T_1 - T_0)$$

$$\lambda = 4\sigma eAT_0^3$$

$$P = \lambda (T_1 - T_0) + \frac{kA(T_1 - T_2)}{\ell} \quad \dots(i)$$

(For left surface)

$$\frac{KA(T_1 - T_2)}{\ell} = \lambda (T_2 - T_0) \quad \dots(ii)$$

(For right surface)

When thickness is doubled

$$P = \lambda (T'_1 - T_0) + \frac{kA(T'_1 - T'_2)}{2\ell} \quad \dots (iii)$$

$$\frac{kA(T'_1 - T'_2)}{2\ell} = \lambda (T'_1 - T_0) \quad \dots(iv)$$

on solving

$$T'_1 = \frac{2T_1^2 + T_1T_2 - T_0(3T_1 - T_2) - T_2^2}{2(T_1 - T_0)} = 333.33k.$$

$$T'_2 = \frac{T_1T_2 + T_0(T_1 - 3T_2) - T_2^2}{2(T_1 - T_0)} = 316.67k.$$

6. Ans (103.04)

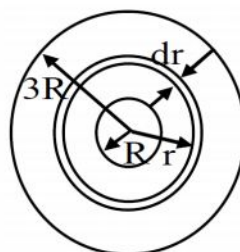
Use doppler effect.

PART-1 : PHYSICS

SECTION-II (ii)

7. Ans (5)

$$V_0 = I R_{eq}$$



$$R_{eq} = \int dR$$

$$R_{eq} = \int \frac{\rho_r dr}{4\pi r^2}$$

$$R_{eq} = \int_R^{3R} \frac{K dr}{4\pi r} \Rightarrow R_{eq} = \frac{K}{4\pi} \ln(3)$$

$$R(r) = \frac{K}{4\pi} \ln\left(\frac{r}{R}\right)$$

$$V(r) = IR(r) = \frac{V_0}{R_{eq}} \times R(r)$$

$$V(2R) = V_0 \frac{\ln 2}{\ln 3}$$

8. Ans (1)

When switch is open

Current in circuit

$$I = \frac{9}{45} A = 0.2A$$

$V_1 + V_2 = 6V$ and because in series same current passes and voltmeters are identical so

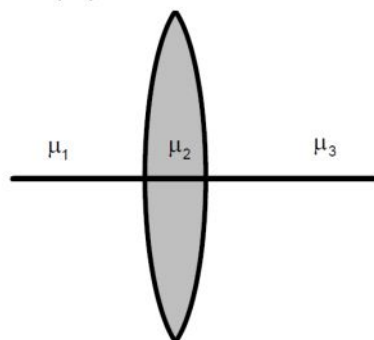
$$V_1 = V_2 = 3V$$

When switch closed

$$V'_1 = 2V; V'_2 = 4V$$

$$\frac{V'_2 - V_2}{V_1 - V'_1} = \frac{4 - 3}{3 - 2} = 1$$

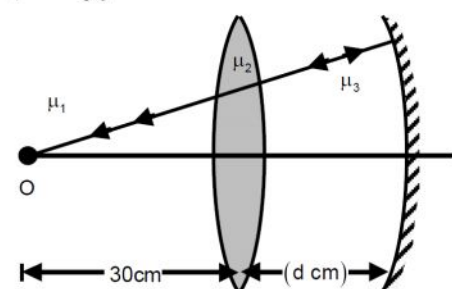
9. Ans (7)



$$\frac{\mu_3}{v} - \frac{\mu_1}{u} = \frac{\mu_3 - \mu_1}{R_1} + \frac{\mu_3 - \mu_2}{R_2}$$

$$\frac{4}{v} - \frac{1}{-30} = \frac{2-1}{10} + \frac{4-2}{-10}$$

$$v = -30$$

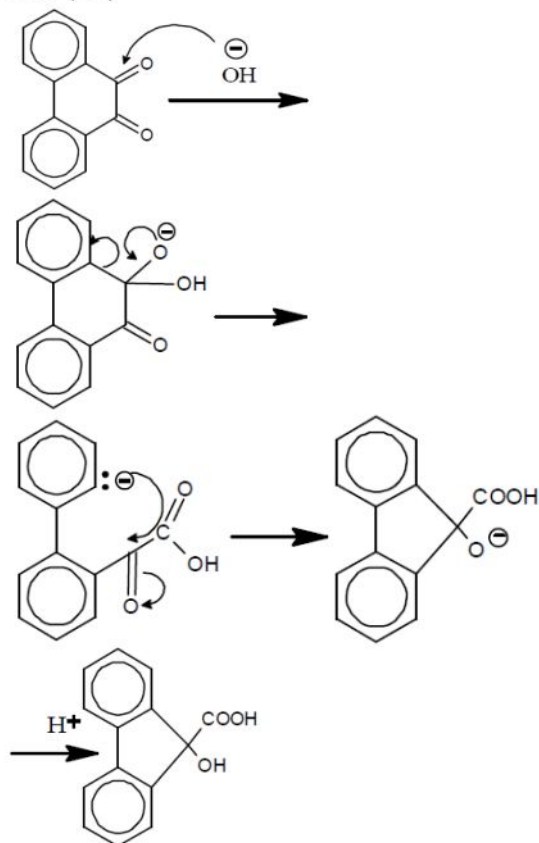


$$d + 30 = 100$$

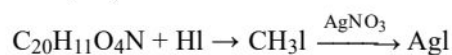
PART-2 : CHEMISTRY

SECTION-I (i)

1. Ans (C)



2. Ans (D)

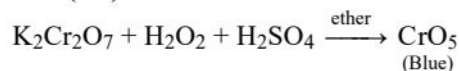


$$\frac{1.24}{329} = .013$$

$$\frac{11.62}{232} = 0.05$$

No. of mole of AgI $\approx 4 \times$ mole of $\text{C}_{20}\text{H}_{11}\text{ON}$

3. Ans (D)



4. Ans (C)

$$\Delta T_f = i K_f \frac{w \times 1000}{\text{M. W.} \times W} = i K_f m$$

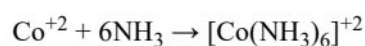
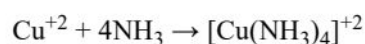
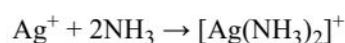
$$.0054 = i \cdot 1.86 \times .001$$

$$i = \frac{.0054}{1.86 \times .001} \approx 2.90$$

PART-2 : CHEMISTRY

SECTION-I (ii)

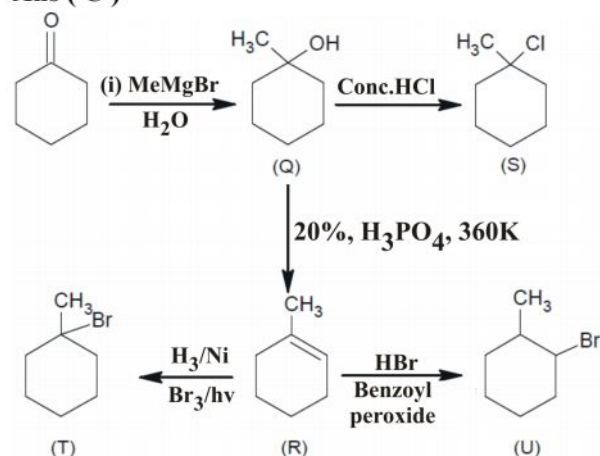
5. Ans (A,B,C)



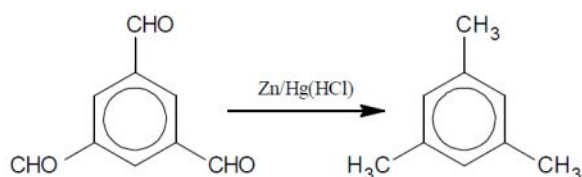
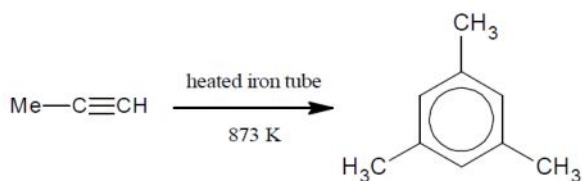
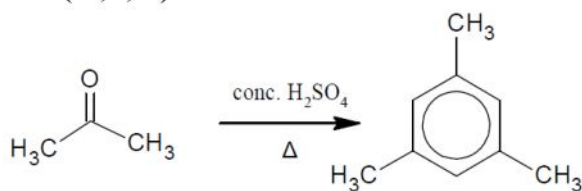
6. Ans (A,B,C)

HNO_3 on dehydration gives N_2O_5

7. Ans (C)



8. Ans (A,B,D)



9. Ans (A,C)

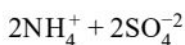
$$\left(\frac{dE}{dT}\right) = \frac{E_2 - E_1}{T_2 - T_1}, \quad \text{According to Nernst}$$

equation, on ↑ in temp. E.M.F. ↓ es.

10. Ans (B,D)



$i = 5$ No. of ions



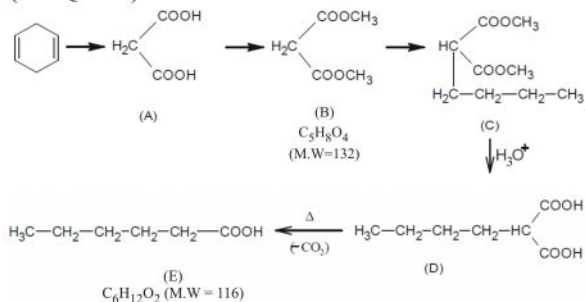
$i = 5 = \text{No. of ions}$

PART-2 : CHEMISTRY

SECTION-II (i)

2. Ans (116.00)

(for Q. 1-2)



4. Ans (3.00)

(for Q. 3-4)

Let the molecular formula of the complex is



∴ % H is in the complex

$$\frac{3x}{52 + 17x + 35.5x} \times 100 = 4.6$$

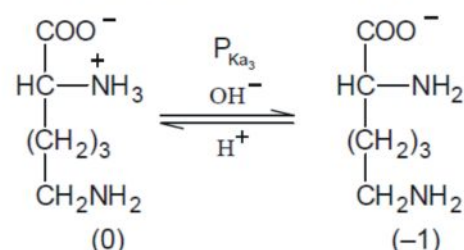
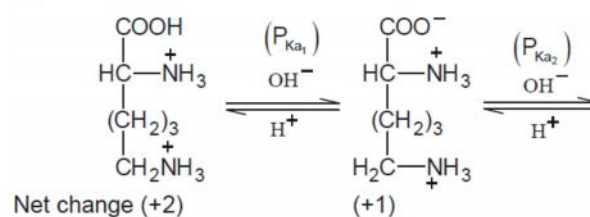
$$x \approx 4.08 \approx 4$$

Now, $\Delta T_f = i k_f m$

$$i = \frac{.00558}{1.86 \times 10^{-3}} \approx 3$$

5. Ans (9.74)

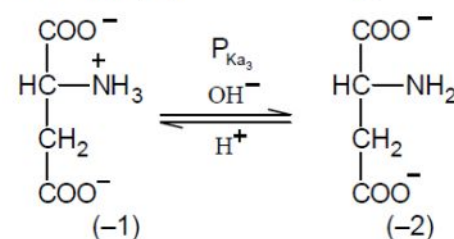
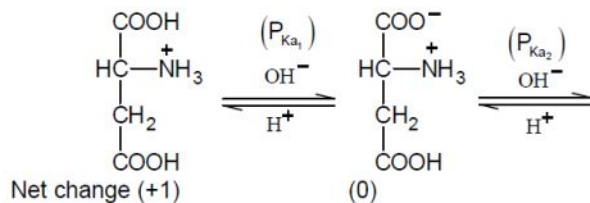
The dissociation of cationic form of lysine can be represented.



$$P_i = \frac{P_{K_{a2}} + P_{K_{a3}}}{2} = \frac{8.95 + 10.53}{2} = 9.74$$

6. Ans (2.77)

The dissociation of cationic form of Aspartic acid can be shown as



$$P_i = \frac{P_{K_{a2}} + P_{K_{a3}}}{2} = \frac{1.88 + 3.65}{2} = 2.77$$

PART-2 : CHEMISTRY

SECTION-II (ii)

7. Ans (7)

$$\frac{1}{\lambda} = R_H(Z)^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

for $n_1 = 1$ $n_2 = ?$

$$\frac{1}{30.4 \times 10^{-9}} = 1.09678 \times 10^7 \times 4 \left[\frac{1}{1^2} - \frac{1}{n_2^2} \right]$$

$n_2 = 2$

for $n_1 = 2$ (first excited state), $n_2 = ?$

$$\frac{1}{108.5 \times 10^{-9}} = 1.09678 \times 10^7 \times 4 \left[\frac{1}{2^2} - \frac{1}{n_2^2} \right]$$

$$\boxed{n_2 = 5}$$

sum of n_1 & $n_2 = 2 + 5 = 7$

8. Ans (6)

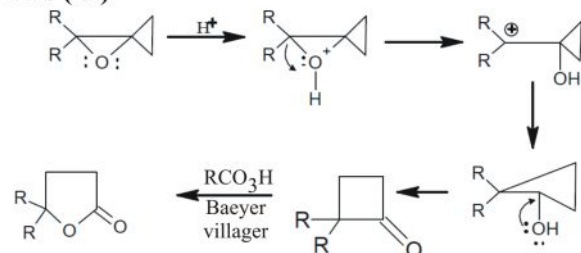
$$\text{Density of AB} = \frac{4 \times M_1}{N_{av} \times \ell_1^3} \quad \dots(i)$$

Let molar mass of AB be M_1 gm/mol and edge length of unit cell be ℓ_1 (m).

$$\text{Density of XY} = \frac{1.3M_1}{N_{av} \times \left(\frac{\ell_1}{2}\right)^3} = \frac{8 \times 3 \times M_1}{N_{av} \times \ell_1^3} \quad \dots(ii)$$

$$(ii)/(i) = 6$$

9. Ans (5)



PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (B)

$$f(x)f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$$

$$\text{let } f(x) = a_n x^n + a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \dots + a_1 x + a_0$$

$$f\left(\frac{1}{x}\right) = \frac{a_n}{x^n} + \frac{a_{n-1}}{x^{n-1}} + a_{n-2} \cdot \frac{1}{x^{n-2}} + \dots + \frac{a_1}{x} + a_0$$

$$f(x) \cdot f\left(\frac{1}{x}\right) = a_0 a_n x^n + (a_{n-1} a_0 + a_n a_1) x^{n-1} + x^{n-2} (a_0 a_{n-2} + a_1 a_{n-1} + a_2 a_n)$$

$$f(x) + f\left(\frac{1}{x}\right) = a_n x^n + a_{n-1} x^{n-1} + x^{n-2} a_{n-2} + \dots + a_1 x + a_0$$

$$+ \frac{a_n}{x^n} + \frac{a_{n-1}}{x^{n-1}} + \frac{a_{n-2}}{x^{n-2}} + \dots + \frac{a_1}{x} + a_0$$

$$a_n = 1 \quad a_0 = 1 ; \{a_1 = a_2 = a_3 = \dots = a_{n-1} = 0\}$$

$$f(x) = x^n + 1$$

$$f(10) = 10^n + 1 = 10^3 + 1$$

$$n = 3$$

$$f(x) = x^3 + 1$$

$$f(20) = (20)^3 + 1 = 8001$$

2. Ans (A)

$$h(x) = \log(f(x) \cdot g(x)) = \log e^{\{y\} + \{y\}} = \{y\} + [y]$$

$$= e^{|x|} \sin x$$

$$\therefore h(x) = e^{|x|} \sin x = \begin{cases} e^x, & x < 0 \\ 0, & x = 0 \\ -e^x, & x > 0 \end{cases}$$

$$\Rightarrow h(-x) = \begin{cases} e^{-x}, & x < 0 \\ 0, & x = 0 \\ -e^{-x}, & x > 0 \end{cases} \Rightarrow h(x) + h(-x) = 0 \text{ for all } x.$$

3. Ans (C)

Let co-ordinates of foot of perpendicular are (a, b, c)

$$\text{Since } \frac{a-11}{10} = \frac{b+2}{-4} = \frac{c+8}{-11} = k$$

$$\text{Then } a = 10k + 11, b = -4k - 2, c = -11k - 8$$

Here all alternative give $a = 1$, then $k = -1$. So foot is

(1,2,3) and perpendicular distance between

$$(1,2,3) \text{ and } (2, -1, 5) \text{ is } \sqrt{14}.$$

4. Ans (B)

Equation of chord of contact

$$xh - yk - 9 = 0 \dots(1)$$

$$x - 9 = 0 \dots(2)$$

$$h = 1 \text{ and } k = 0$$

pair of tangent

$$SS' = T^2$$

$$(x^2 - y^2 - 9)(-8) = (x - 9)^2 9x^2 - 8y^2 - 18x + 9 = 0$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. Ans (A,B,D)

$$(A) (1 + x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + {}^nC_3x^3 + \dots + {}^nC_nx^n$$

$$\text{Put } x = \omega \text{ where } \omega = -\frac{1}{2} + \frac{i\sqrt{3}}{2}$$

$$({}^nC_0 + {}^nC_3 + {}^nC_6 + \dots) + \dots + ({}^nC_1 + {}^nC_4 + {}^nC_7 + \dots)\omega + ({}^nC_2 + {}^nC_5 + {}^nC_8 + \dots)\omega^2 = (1 + \omega)^n$$

$$({}^nC_0 + {}^nC_3 + {}^nC_6 + \dots) + ({}^nC_1 + {}^nC_4 + {}^nC_7 + \dots)$$

$$\left(-\frac{1}{2} + \frac{i\sqrt{3}}{2}\right) + ({}^nC_2 + {}^nC_5 + {}^nC_8 + \dots)$$

$$\left(-\frac{1}{2} + \frac{i\sqrt{3}}{2}\right)$$

$$({}^nC_0 + {}^nC_3 + {}^nC_6 + \dots) - \frac{1}{2} ({}^nC_1 + {}^nC_2 + {}^nC_4 + {}^nC_5 + \dots) + \frac{i\sqrt{3}}{2} ({}^nC_1 - {}^nC_2 + {}^nC_4 - {}^nC_5 + \dots) = (-\omega^2)^n$$

$$(B) a^2b^2 = 4 \Rightarrow 5\log_2(a) + 2\log_2(b) = 2$$

$$y = 10\log_2a \cdot 4\log_2b = 40\log_2a \cdot \log_2b$$

$$A.M. \geq G.M. \Rightarrow \frac{5\log_2(a) + 2\log_2(b)}{2} \geq \sqrt{10\log_2a \cdot \log_2b}$$

$$\log_2a \cdot \log_2b \leq \frac{1}{10} \Rightarrow y \leq 4, y = 4 \text{ when } 5\log_2a = 2\log_2b \Rightarrow a^5 = b^2 = 2$$

$$(C) \text{ For constant term put } x = 0 \text{ and get constant term} = 4$$

$$(D) \text{ Coefficient of } x^{24} = 1.25 + 2.24 + 3.23 + \dots + 25.1 = 2925$$

6. Ans (A,B,C)

Let R be the common ratio of the G.P. and D be the

common difference of A.P.

$$a_5 = a_5, a_9 = Ra_5, a_{16} = a_5R^2$$

$$a_9 - a_5 = 4D \Rightarrow (R - 1)a_5 = 4D$$

$$a_{16} - a_9 = 7D \Rightarrow R(R - 1)a_5 = 7D$$

$$\text{From equation (1) / (2), we get } \frac{1}{R} = \frac{4}{7} \Rightarrow R = \frac{7}{4}$$

$$\text{From equation (2) - (1), we get } (R - 1)^2 a_5 = 3D$$

$$\Rightarrow \frac{9a_5}{16} = 3D$$

$$\frac{3}{16}(a_1 + 4D) = D \Rightarrow \frac{3}{16}a_1 = \left(1 - \frac{3}{4}\right)D \Rightarrow a_1 = \frac{4D}{3}$$

7. Ans (A,D)

$$\frac{2x}{(x-1)(x-4)} = \frac{C}{x-1} + \frac{D}{x-4}$$

$$2x = C(x-4) + D(x-1)$$

$$\therefore C = -2/3, D = 8/3$$

$$\therefore \int \frac{e^{x-1}}{(x-1)(x-4)} 2x \, dx = \int e^{x-1} \left(\frac{-2/3}{x-1} + \frac{8/3}{x-4} \right) dx$$

$$= -\frac{2}{3} F(x-1) + \frac{8}{3} e^3 F(x-4) + C$$

$$\therefore A = -2/3, B = 8/3 e^3$$

8. Ans (B,C,D)

Let $\hat{i}, \hat{j}, \hat{k}$ be the unit vectors along OA, OB, OC.

CN, BM are diagonals skew to OA

Let l be the S.D between OA and CN and let

$$\vec{OA} = x\hat{i}, \vec{OB} = y\hat{j}, \vec{OC} = z\hat{k} \Rightarrow \vec{ON} = x\hat{i} + y\hat{j}$$

Equation of OA and CN are $\vec{r} = x\hat{i}$ and

$$\vec{r} = z\hat{k} + p(x\hat{i} + y\hat{j} - z\hat{k}) \text{ respectively.}$$

∴ The shortest distance between those two lines.

$$l = \frac{yz}{\sqrt{y^2 + z^2}} \Rightarrow \frac{1}{l^2} = \frac{1}{y^2} + \frac{1}{z^2}$$

$$\text{Similarly } \frac{1}{m^2} = \frac{1}{z^2} + \frac{1}{x^2}, \frac{1}{n^2} = \frac{1}{x^2} + \frac{1}{y^2},$$

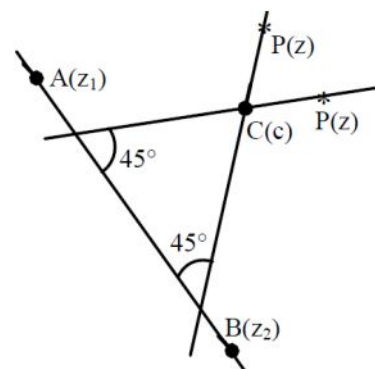
$$\Rightarrow \frac{1}{l^2} + \frac{1}{m^2} + \frac{1}{n^2} = 2 \left(\frac{1}{OA^2} + \frac{1}{OB^2} + \frac{1}{OC^2} \right)$$

9. Ans (A,B,C)

Let P(z) be any point on the required line.

Then, $\frac{\vec{CP}}{|\vec{CP}|}$ i.e. $\frac{z-c}{|z-c|}$ is a unit vector parallel to

it



Let A(z₁) and B(z₂) be two points on

$\bar{a}z + a\bar{z} + b = 0$ then $\frac{z_2 - z_1}{|z_2 - z_1|}$ is a unit vector

parallel to the line

$$\bar{a}z + a\bar{z} + b = 0$$

$$\begin{aligned} \frac{z-c}{|z-c|} &= \frac{z_2 - z_1}{|z_2 - z_1|} e^{\pm i\left(\frac{\pi}{4}\right)} \\ \frac{(z-c)^2}{(z-c)(\bar{z}-\bar{c})} &= \frac{(z_2 - z_1)}{(z_2 - z_1)(\bar{z}_2 - \bar{z}_1)} e^{\pm i\frac{\pi}{2}} \\ \frac{z-c}{z-\bar{c}} &= \pm i \frac{(z_2 - z_1)}{(z_2 - \bar{z}_1)} \end{aligned}$$

∴ A(z₁) and B(z₂) are on the line

$$\bar{a}z + a\bar{z} + b = 0 \text{ therefore } \bar{a}z_1 + a\bar{z}_1 + b = 0$$

$$\bar{a}z_2 + a\bar{z}_2 + b = 0$$

$$\Rightarrow -\frac{a}{\bar{a}} = \frac{z_2 - z_1}{\bar{z}_2 - \bar{z}_1}$$

From equation (1) and (2) we get

$$\frac{z-c}{a} \pm \frac{i(\bar{z}-\bar{c})}{\bar{a}} = 0$$

10. Ans (A,B,C,D)

$$f(x) = x^{1/3} (x - 1)$$

$$\Rightarrow \frac{df(x)}{dx} = \frac{4}{3}x^{1/3} - \frac{1}{3} \cdot \frac{1}{x^{2/3}} = \frac{1}{3x^{2/3}} [4x - 1]$$

$\Rightarrow f(x)$ changes sign from -ve to +ve, $x = 1/4$,

which is point of minima

Also $f(x)$ does not exist at $x = 0$ as $f(x)$ has vertical tangent at $x = 0$.

$$f''(x) = \frac{4}{9} \cdot \frac{1}{x^{2/3}} = \frac{1}{3} \cdot \frac{2}{3} \cdot \frac{1}{x^{5/3}} = \frac{2}{9x^{2/3}} \left[2 + \frac{1}{x} \right] = \frac{2}{9x^{2/3}} \left[\frac{2x+1}{x} \right]$$

$\therefore f''(x) = 0$ at $x = -\frac{1}{2}$ which is the point of

inflection at $x = 0$, $f''(x)$ does not exist but $f'(x)$

changes sign, hence $x = 0$ is also the point of inflection.

From the information the graph of $y = f(x)$ is as shown.

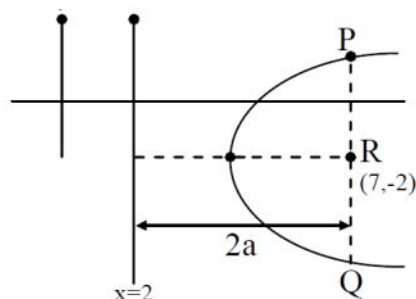
PART-3 : MATHEMATICS

SECTION-II (i)

1. Ans (25.00)

$$(x-2) = \sqrt{(x-7)^2 + (y+2)^2}$$

It is parabola



$$(PR)(RQ) = 4a^2$$

$$= 4 \times \left(\frac{5}{2}\right)^2$$

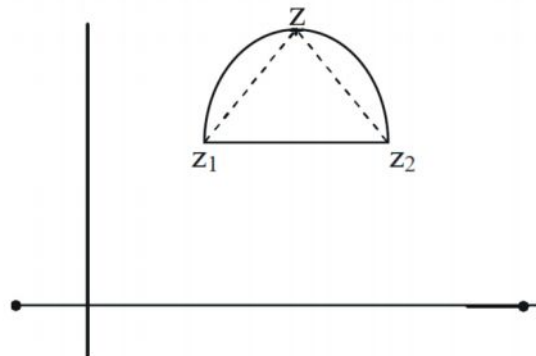
$$= 25$$

2. Ans (1.57)

$$\operatorname{Re} z - 2 = |z - 7 + 2i|$$

Let $P(z_1), Q(z_2)$

$$\arg\left(\frac{z_1 - (2 + \alpha i)}{z_2 - (2 + \alpha i)}\right) = \frac{\pi}{2} (\alpha \in \mathbb{R})$$



3. Ans (0.86)

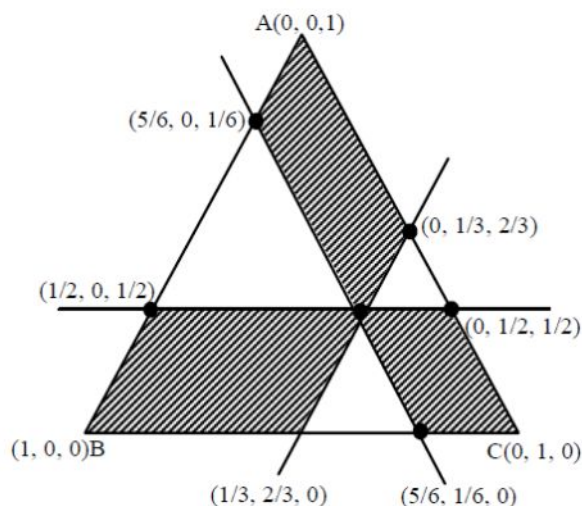
T is an equilateral triangle with the vertices at (1, 0, 0), (0, 1, 0) and (0, 0, 1)

$\Rightarrow$ area of the region T is $\frac{\sqrt{3}}{2}$.

4. Ans (0.34)

Take a point $P\left(\frac{1}{2}, \frac{1}{3}, \frac{1}{6}\right)$ on the plane

$$x + y + z = 1$$



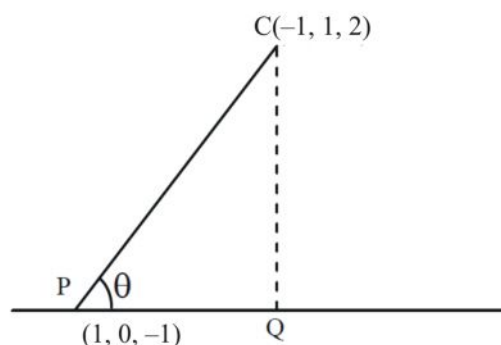
The region S is shown as shaded region.

$$\text{Area of } S = \frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{4} (a^2 + b^2 + c^2), \text{ where } a, b, c$$

$$\text{are sides of the small equilateral triangles} = \frac{7\sqrt{3}}{36}.$$

6. Ans (0.45)

$$\frac{x-2}{z} = \frac{y}{-1} = \frac{z+1}{2}$$



$$CQ = CP \sin \theta$$

$$= \frac{|\vec{CP} \times \vec{L}|}{|\vec{L}|}$$

$$= \frac{\left\| \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -1 & -3 \\ 2 & -1 & 2 \end{vmatrix} \right\|}{\sqrt{4+1+4}} = \frac{\sqrt{125}}{3}$$

PART-3 : MATHEMATICS

SECTION-II (ii)

7. Ans (2)

E = even of any one cutting a space in one cut

$$n(E) = {}^{13}C_1$$

$$n(S) = {}^{52}C_1$$

$$P(E) = 1/4 = p, P(\bar{E}) = q$$

Probability of a winning = $p + qqpp + qqqqpp$

+ ∞

$$= \frac{p}{1-q^3} = \frac{64}{175} = 128$$

$$\left[\frac{\lambda}{64} \right] = 2$$

8. Ans (8)

$$f(x) = \begin{cases} ax^2 + bx & \text{for } -1 < x < 1 \\ \frac{a-b-1}{2} & x = -1 \\ \frac{a+b+1}{2} & x = 1 \\ \frac{1}{x} & \text{for } x > 1 \text{ or } x < -1 \end{cases}$$

for continuity at $x = 1$ we have $a + b = \frac{a+b+1}{2}$

$$\text{hence, } a + b = 1 \quad \dots(1)$$

for continuity at $x = -1$

$$a - b = -1 \quad a - b = -1 \quad \dots(2)$$

hence $a = 0$ and $b = 1$

9. Ans (7)

$$P(A \cap B) = P(A)P(B) \Rightarrow 1/6 = P(A)P(B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\Rightarrow 2/3 = P(A) + \frac{1}{6P(A)} - \frac{1}{6}$$

$$\Rightarrow 6(P(A))^2 - 5P(A) + 1 = 0 \Rightarrow P(A) = 1/2, P(B)$$

$$= 1/3$$

$$\text{So, } 8P(A) + 9P(B) = 4 + 3 = 7$$